

/ NXP NAFE13388-UIM User Guide



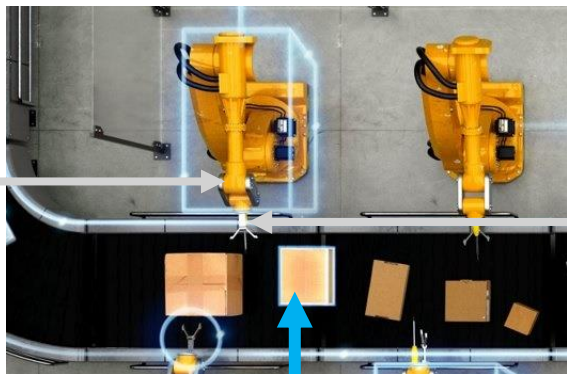
03.16.2026 **Kami Huang**, Staff FAE



/AFE and EP enable the smart factory



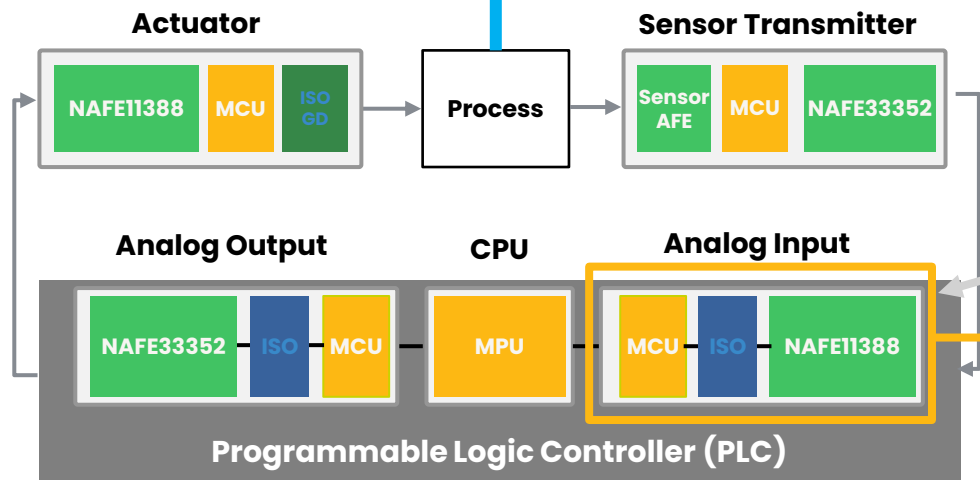
Actuator



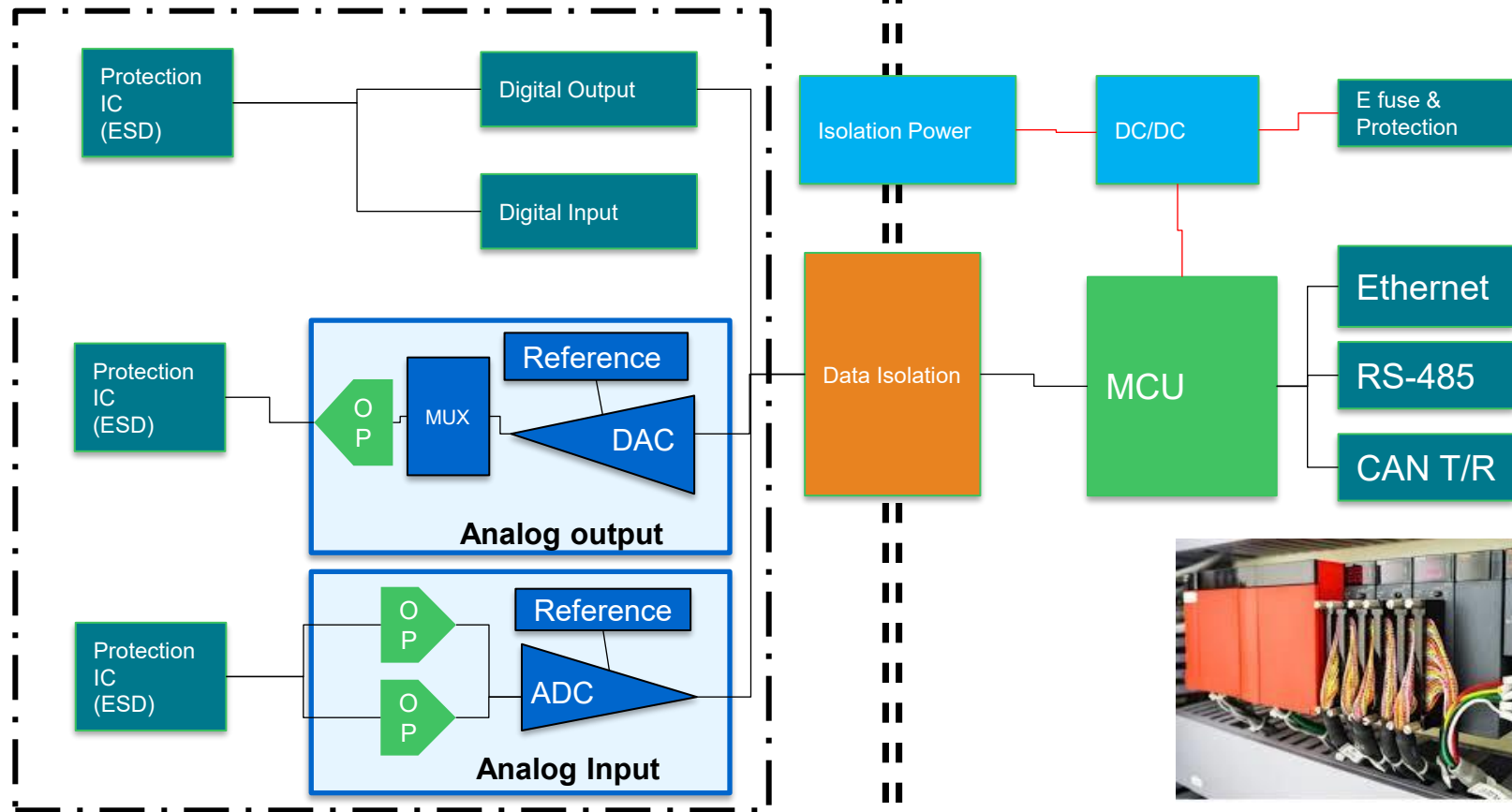
Sensor
Transmitter

Long
wires

Long
wires

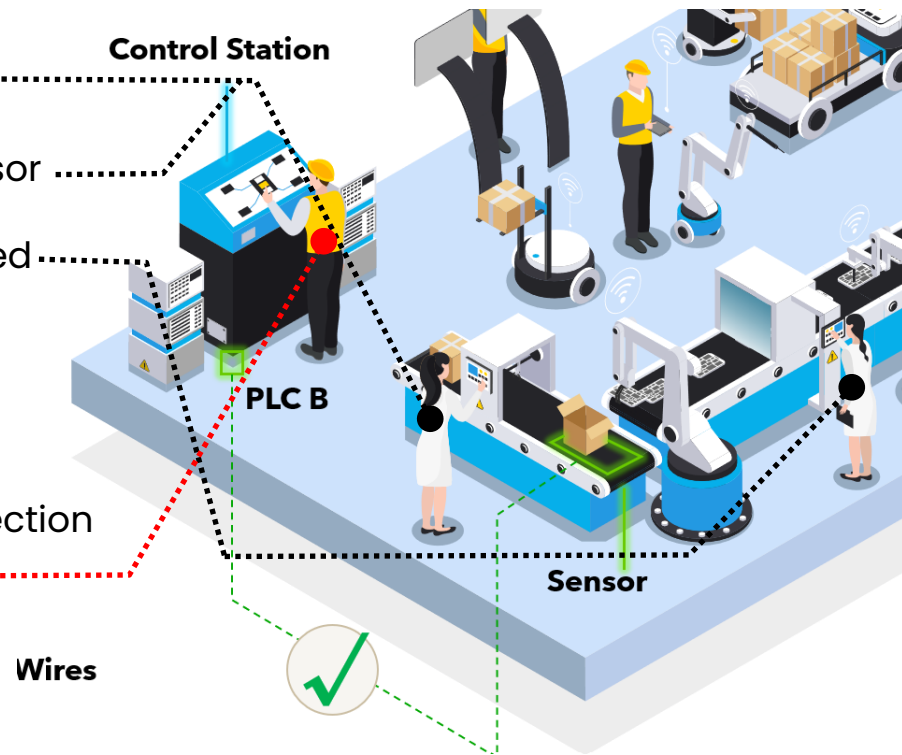


PLC



- ✓ PLC I/O is connected to the sensor
- ✓ I/O Module is hardware compatible with the sensor
- ✓ The AFE inside the I/O Module has been configured

? How to verify that the configuration and the connection has been properly done





NAFE11388 / NAFE71388

NAFE13388 / NAFE73388

Highly Configurable 8 Channels ± 25 V
Universal Input Analog Front-End
with and without Excitation Source

NXP Industrial AFEs

WHY NAFE?

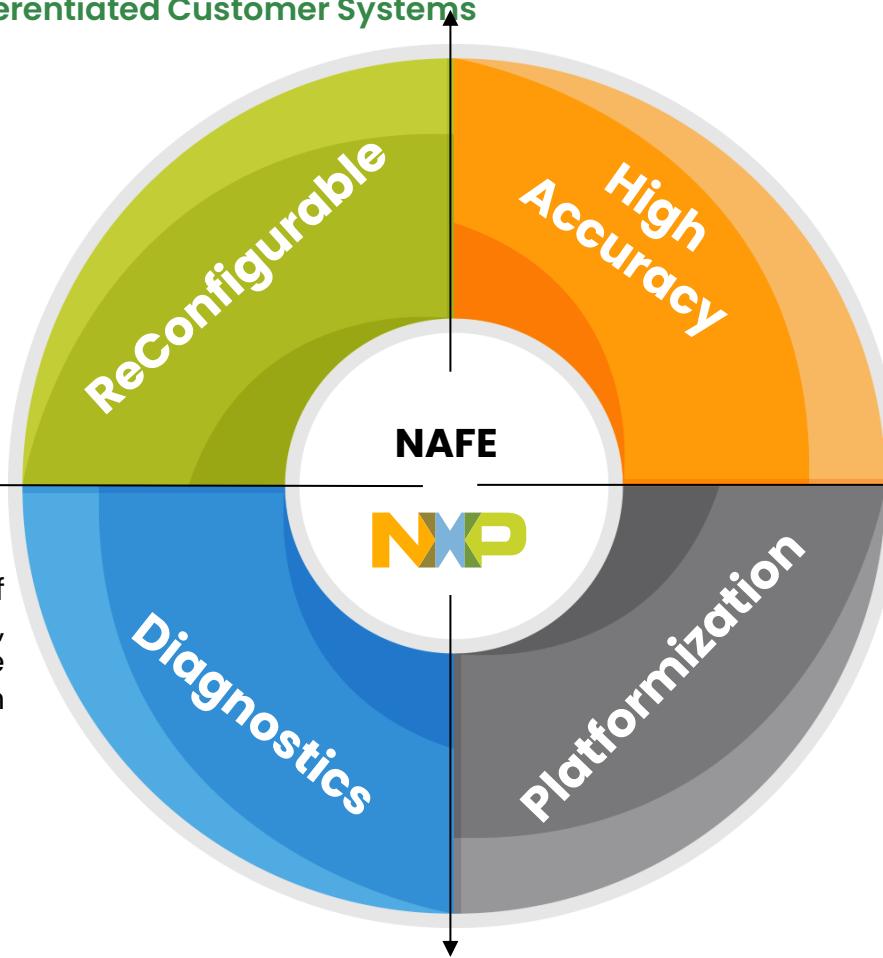
Unique AFE Family for Differentiated Customer Systems



Fully software-configurable
universal inputs

Measures various sensor
inputs without hardware
changes

Easy implementation of
predictive maintenance,
anomaly detection, failure
prediction

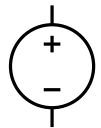


$\pm 0.01\%$ accuracy at room and
 $\pm 0.05\%$ accuracy over
temperature

17-bit ENOB at 144kSPS
(NAFE7x388) and at 72kSPS
(NAFE1x388)

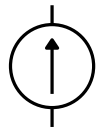
Complete family of products,
with or without Factory
Calibration, 16 or 24 bits reduce
the development and
qualification effort and time by
using a single platform for a
range of products

Universal Input Configuration Challenges



VOLTAGE
0-10 V

requires high input voltage range up to 12.5 V



Current
0-20 mA

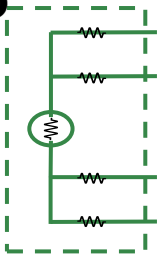
Requires reading currents from 0 to 20 mA with 0.2 % accuracy FS. $R_{sense} = 50\ \Omega$
Minimum voltage level 2mV
12 bits resolution is enough

TC



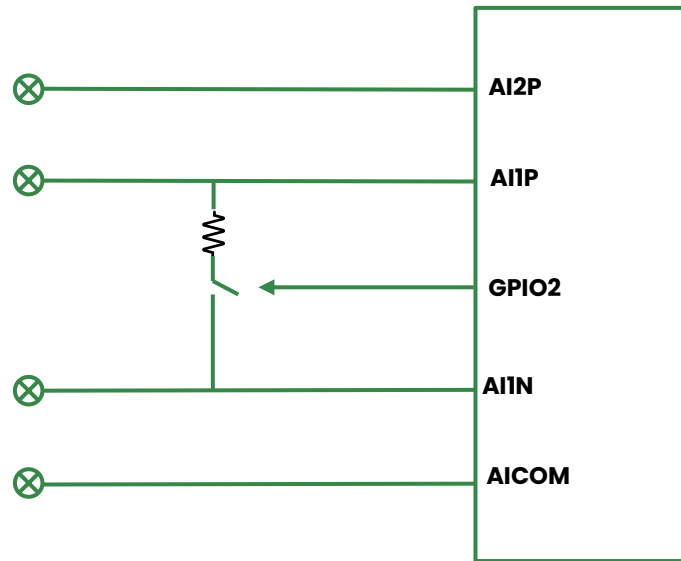
Example: K-Type
-270 to 1370 °C
16 bits resolution is enough
1 °C Accuracy Needs Noise < 2μVRMS

RTD



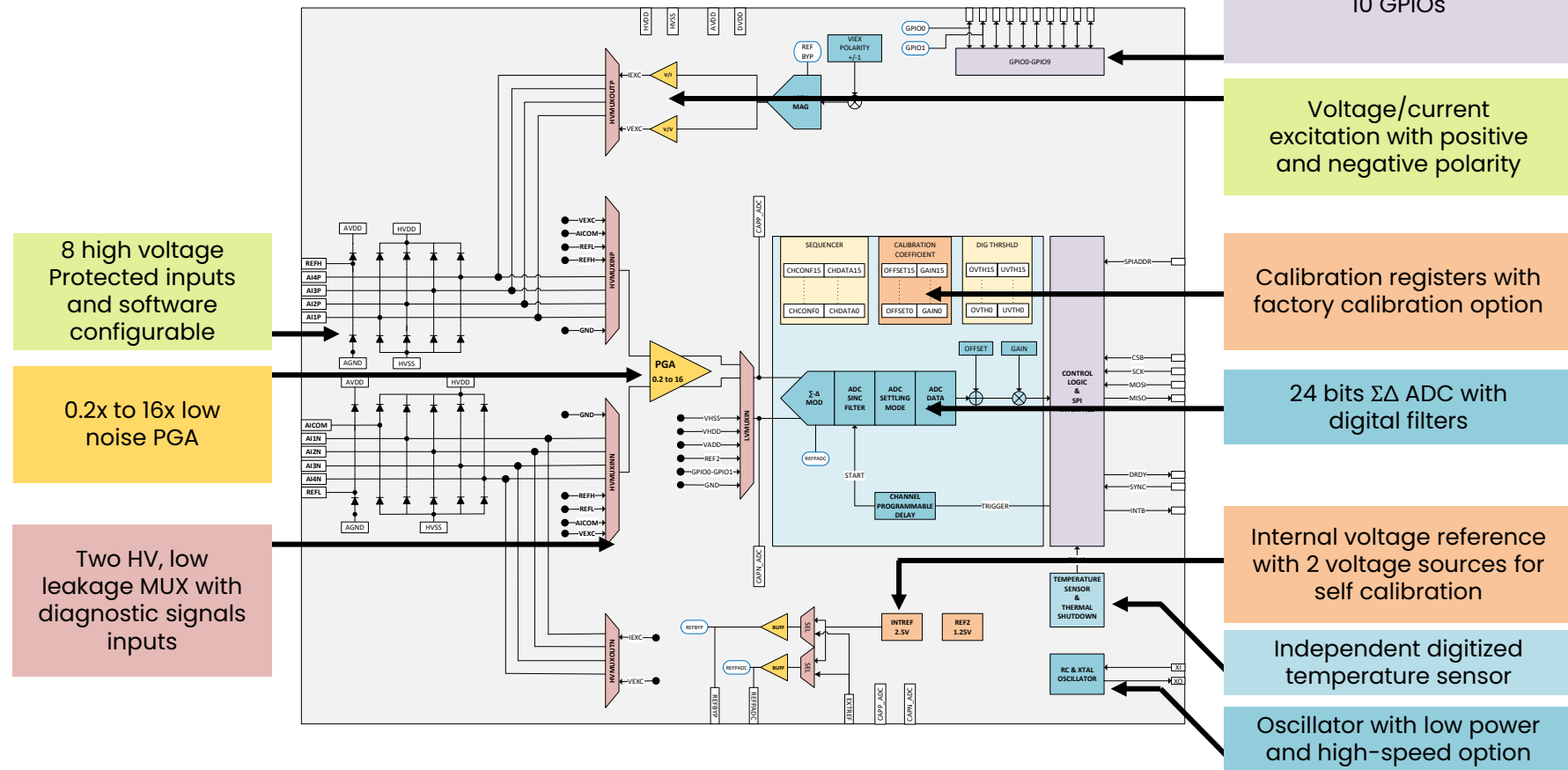
Pt100: -200 to 800 °C
100 μA excitation current
0.5 °C accuracy requires 11 bits but Noise < 22 μVrms
0.1 °C accuracy requires 13 bits but Noise < 4 μVrms

CHALLENGE: 10 V and 2 μVRMS Noise need to be supported by the signal chain



180 db
Dynamic Range

NAFE Series: What's inside



NAFE in Numbers

Distinctive features

8 configurable HV inputs

- Single-ended or differential, with ranges up to +/-25 V
- Configurable for voltage, current, resistance, RTD, Thermocouples
- **Over-voltage protected up to +/-36 V**

Fast Data Rate

- **NAFE11388: 7.5 SPS to 288 kSPS**
- **NAFE71388: 15SPS to 576 kSPS**
- Simultaneous 50Hz/60 Hz line rejection
- **Fastest eight-channel read time = 240 us30 us/channel**

Excitation Source

- **NAFE13388 / NAFE73388**

High Accuracy

- ± 0.01 % accuracy at room
- **± 0.05 % accuracy over temperature**
- ENOB:
 - **NAFE11388: 17-bit at 72 kSPS / 24-bit at 30 SPS**
 - **NAFE71388: 17-bit at 144 kSPS / 24-bit at 50 SPS**

System Calibration

- End-to-end calibration with integrated precise voltage sources
- Accurate digitally calibrated products are available

- Full readout ± 3 °C accurate temperature sensor
- 0.2 % accuracy for internal oscillator
- Diagnostic system for faults detection and prediction
- CRC error detection
- 10 GPIOs
- 30 MHz SPI Interface
- **7.5 kV HBM ESD and IEC61000-4-5 2 kV Surge protection**
- Power supply: HV: ± 7 V to ± 24 V, LV: 3.3 V
- Operating temperature range TA: -40 °C to 125 °C
- Package: 64 pin, 9 mm x 9 mm x 0.85 mm VQFN



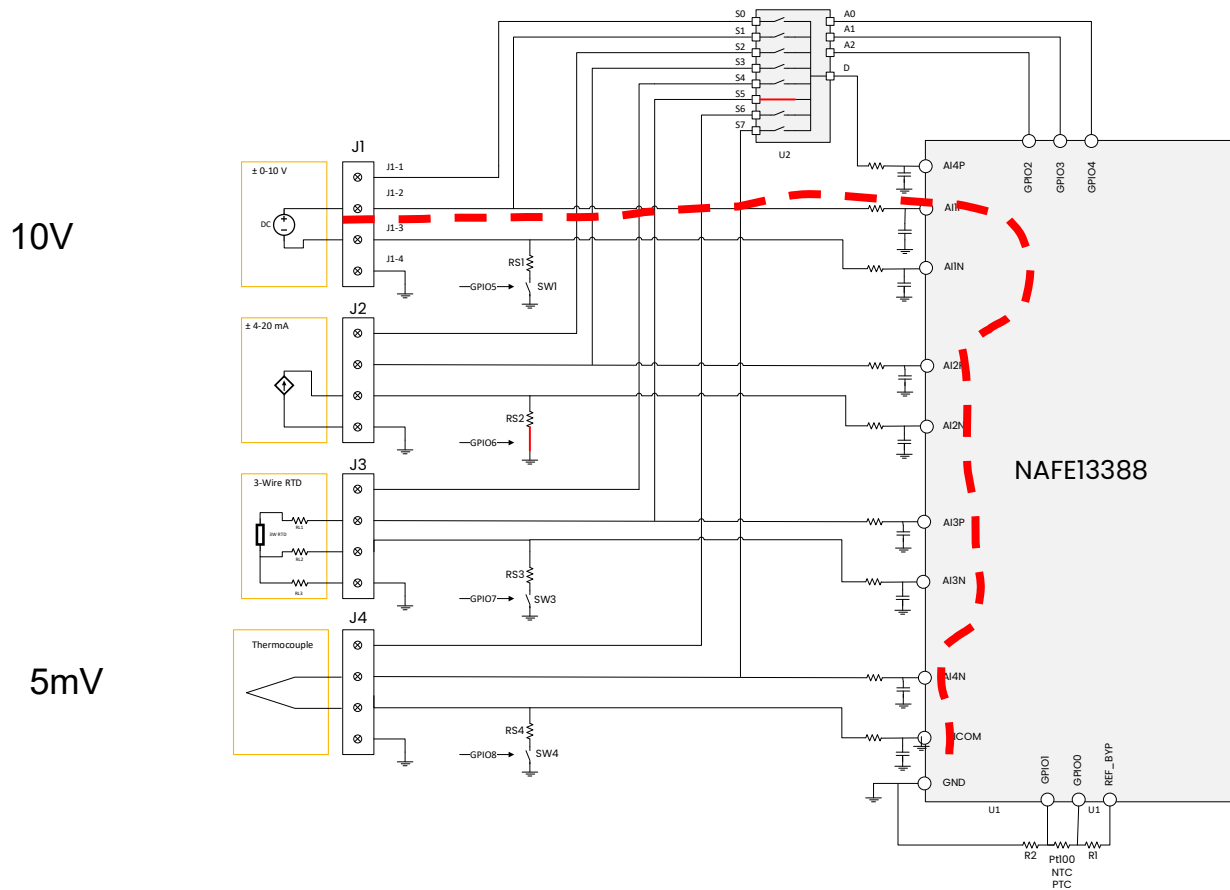
Product **Longevity**

Product Longevity |
NXP Semiconductors

THE GOAL OF NXP'S PRODUCT LONGEVITY PROGRAM:

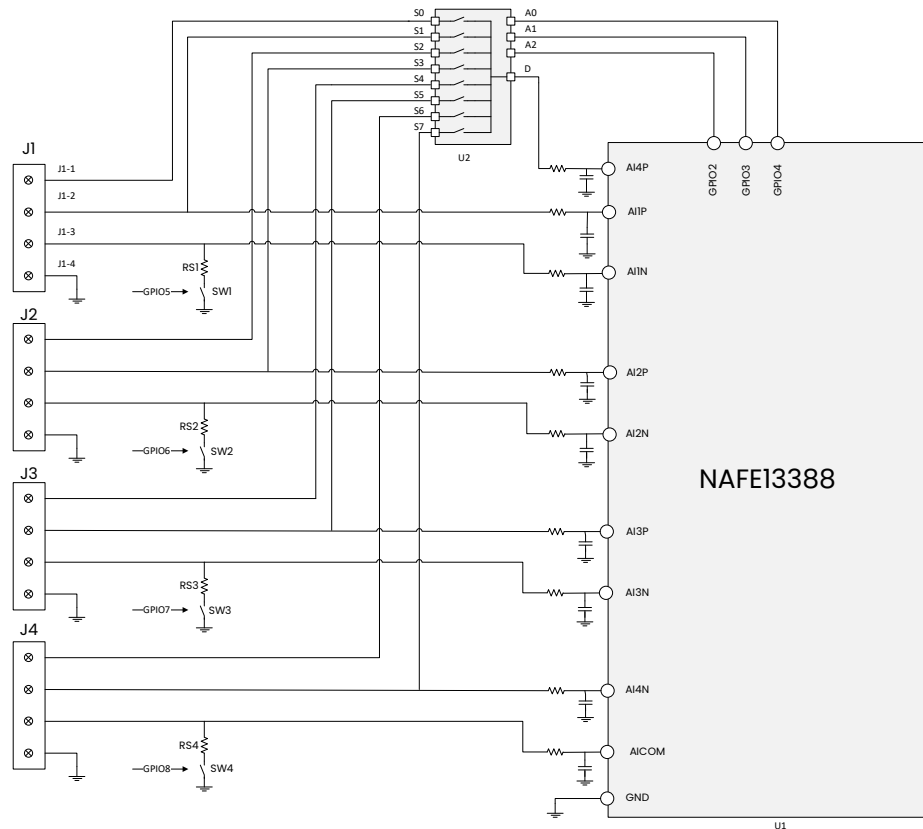
TO ENSURE A STABLE SUPPLY
OF PRODUCTS FOR YOUR
EMBEDDED DESIGNS

What is the Cross talk? Why is important



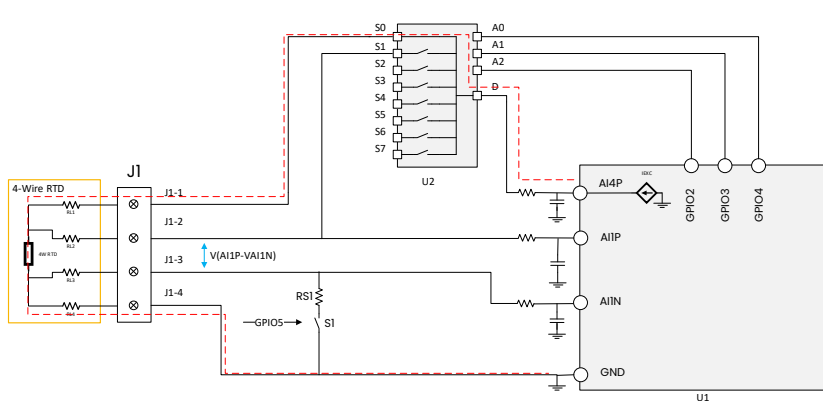
4 Channels Universal Input Schematic

	Voltage Input	Current Input	4-wire RTD	3-Wire RTD	2-wire RTD	Thermocouple	Wheatstone
J1	AI1P-AI1N	AI1N-GND	Force: S0-GND Sense: AI1P-AI1N	Force: S1-GND Sense: AI1P-AI1N	Force/Sense: AI1P-AI1N	AIP-AIN CJC: GPIO0-1	Force: S0-GND Sense: AI1P-AI1N
J2	AI2P-AI2N	AI2N-GND	Force: S2-GND Sense: AI2P-AI2N	Force: S3-GND Sense: AI2P-AI2N	AI2P-AI2N	AI2P-AI2N CJC: GPIO0-1	Force: S2-GND Sense: AI2P-AI2N
J3	AI3P-AI3N	AI3N-GND	Force: S4-GND Sense: AI3P-AI3N	Force: S5-GND Sense: AI3P-AI3N	AI3P-AI3N	AI3P-AI3N CJC: GPIO0-1	Force: S4-GND Sense: AI3P-AI3N
J4	AI4N-AICOM	AICOM-GND	Force: S6-GND Sense: AI4N-AICOM	Force: S7-GND Sense: AI4N-AICOM	AI4N-AICOM	AI4N-AICOM CJC: GPIO0-1	Force: S6-GND Sense: AI4N-AICOM



4 and 3-Wire RTD

The 4-wire RTD measurement is implemented by injecting the current using the integrated current source in the NAFE13388. The current is routed to J1-1 by closing the S0 switch of U2. It goes through the RTD and returns to ground. The figure shows the resulting current flow. The equivalent series resistances, RL1 to RL4, of the RTD wires are also shown. Since the current does not flow through the sensing wires, the resulting voltage of VAI1P-VAI1N provides the measurement with the required accuracy.



For the 3-wire RTD measurement, the current is injected by the integrated current source of the NAFE13388.

The current is routed to J1-2 by closing the S1 switch of U2. It goes through the RTD and returns to ground.

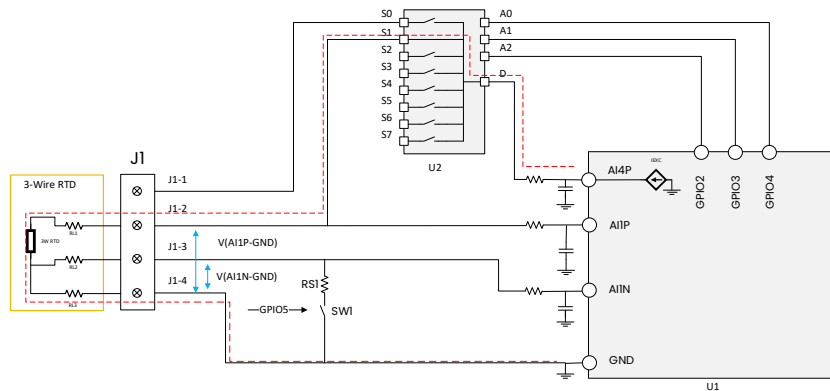
The Figure shows the resulting current flow. The equivalent series resistances, RL1 to RL3, of the RTD wires are also shown.

The resulting voltage (V1) is measured between J1-2 and GND. Since the wire resistances RL1 and RL3 are also part of the current flow, the measured voltage includes also the wire resistances contribution.

The NAFE13388 reduces the error caused by this wire resistance by measuring also the voltage (V2) between J1-3 and GND which measures the voltage drop caused by RL3 only.

The RTD resistance will be then calculated as:

$$R_{RTD} = (V1 - 2 \cdot V2) / I_{EXC}$$

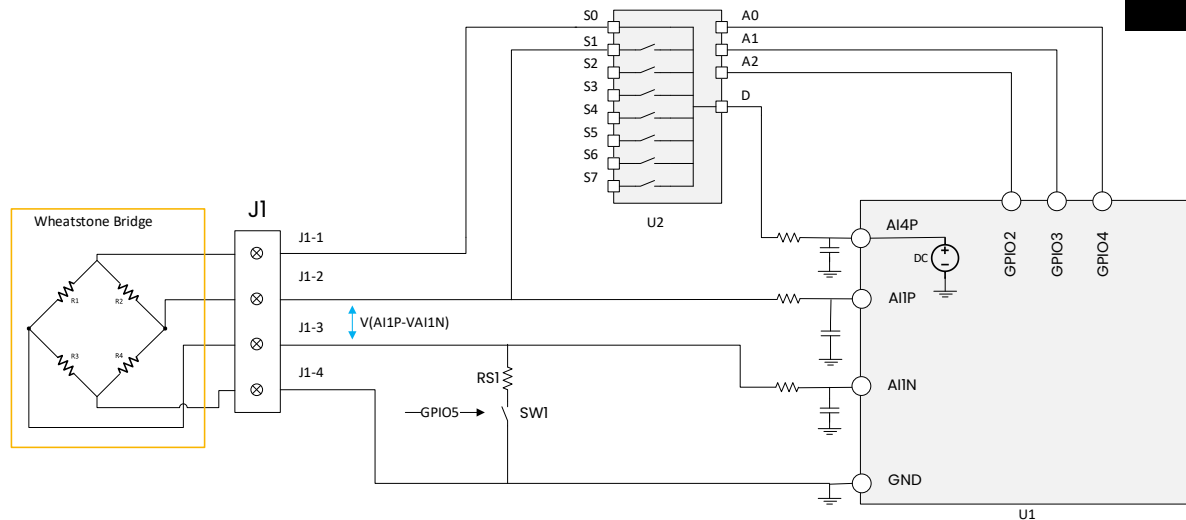


Wheatstone Bridge

The NAFE13388 excitation source can also be configured as voltage output. In this way, it is possible to support a Wheatstone bridge measurement without additional components.

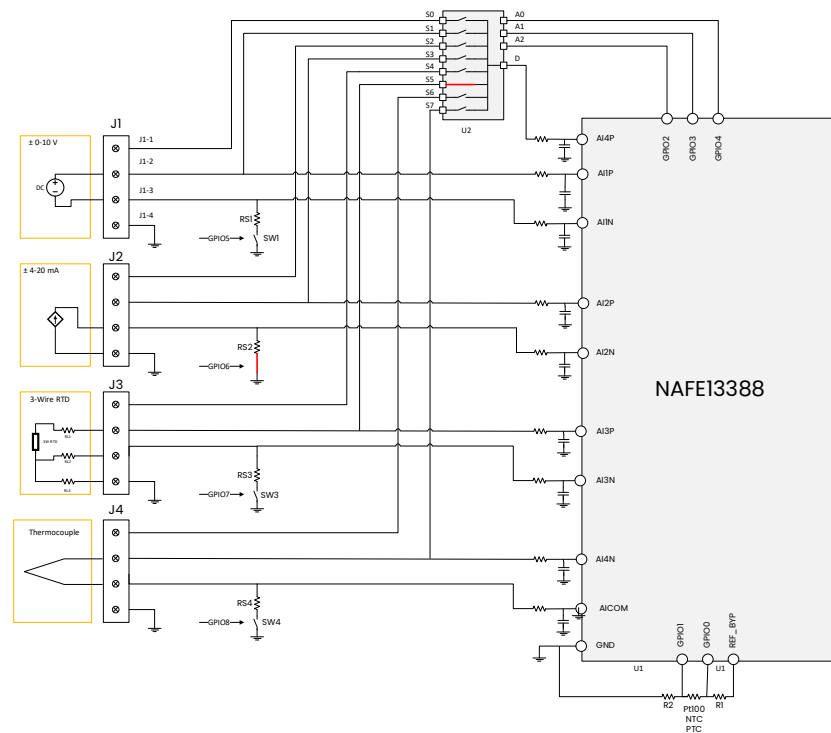
When configured as voltage output, the excitation source has an output impedance of 500 Ω . The applied voltage at J1-1 is the result of the excitation impedance and bridge impedance that work as a resistor divider.

In case the measured Wheatstone bridge has low impedance, a voltage buffer might be needed. The bridge output is measured differentially between AI1P and AI1N.



4 Ch Universal Input Solution Test Results

	Input Type	Logical Channel	IN+	IN-	Excitation	PGA	Datarate	Digital Filter
J1	Voltage Input	LCH0	AI1P	AI1N	OFF	0.2x	1 kSPS	SINC4+2
J2	Current Input	LCH1	GND	AI2N	OFF	0.4x	1 kSPS	SINC4+2
J3	RTD: 3-wire Pt100	LCH2 LCH3	AI3P GND	AI3N AI3N	S5 S5	16x	50 SPS	SINC4+3
J4	Thermocouple with CJC	LCH4	AICOM	AI4N	OFF	16x	50SPS	SINC4+3
	Pt100 fo CJC	LCH5	GPIO0	GPIO1	OFF	-	50 SPS	SINC4+3



Function	LCH	IN+	IN-	PGA	Data rate	SINC Filter	VIEXC	DMM	MEAS	Error [%]
VI	0	AI1P	AI1N	0.2	1k	SINC4+2	OFF	6.995 V	6.996 V	0.017%
CI	1	AI2P	GND	0.4	1k	SINC4+2	OFF	14.211 mA	14.216 mA	0.036%
3w RTD	2	AI3P	GND	16	50	SINC4+3	S5		0.050836 V	
	3	AI3N	GND	16	50	SINC4+3	S5		0.000504 V	
		$(V_{LCH2} - 2 * V_{LCH3}) / I_{EXC}$						99.690 Ω	99.656 Ω	-0.034%
TC	4	AI4N	AICOM	16	50	SINC4+3	OFF	0.005160 V	0.005156 V	-0.078%
CJC	5	GPIO0	GPIO1	x	50	SINC4+3	OFF	0.04541 V	0.04540 V	-0.010%

AIO AFE By Numbers

Distinctive features

- **One analog Input/Output and two universal input** software configurable channels
- Single ended, differential, and multi-ranges **software configurable** inputs and output
 - voltage, current, resistance, RTD, Thermocouples
 - I/O pins protected up to $\pm 36V$ and up to $\pm 60V$ with limiting resistors
- **Output Characteristics**
 - 14, 16, 18 bits DAC options
 - $\pm 12.5 V$ output,
 - $\pm 25 mA$ output with 50Ω external resistor
 - $10\mu s$ settling time
- **Input Characteristics**
 - $\pm 25 V$ to $\pm 750 mV$ input, 24-bit $\Sigma\Delta$ ADC
 - $\pm 0.01\%$ accuracy at room, $\pm 0.05\%$ from $0 ^\circ C$ to $+60 ^\circ C$
 - ENOB: 17-bit at 72 kSPS / 24-bit at 30 SPS
 - 50/60Hz Normal Mode rejection
- **System Calibration**
 - End-to-end calibration with integrated precise voltage source
 - Accurate digitally calibrated products are available

- Full readout $\pm 3 ^\circ C$ accurate temperature sensor
- 0.2% accuracy for internal oscillator
- 5 GPIOs with dual function
- Diagnostic system for faults detection and prediction
- CRC error detection
- 30MHz SPI Interface
- Tested for IEC61000-4-5 surge up to $\pm 2 kV$
- Tested for IEC61000-4-4 Electrical Fast Transient and burst immunity up to $\pm 2.5 kV$ on Analog Input and Output pins
- Power supply: HV: $\pm 7 V$ to $\pm 24 V$, LV: $3.3 V$
- Operating temperature range T_A : $-40 ^\circ C$ to $125 ^\circ C$
- Package: 40 pin, 6mm x 6mm


Product Longevity



Product Longevity |
NXP Semiconductors

THE GOAL OF NXP'S PRODUCT LONGEVITY PROGRAM:

TO ENSURE A STABLE SUPPLY
OF PRODUCTS FOR YOUR
EMBEDDED DESIGNS

DEVELOPMENT ECOSYSTEM

Released

AVNET

- **NAFE13388-EVB**
NAFE11388-EVB

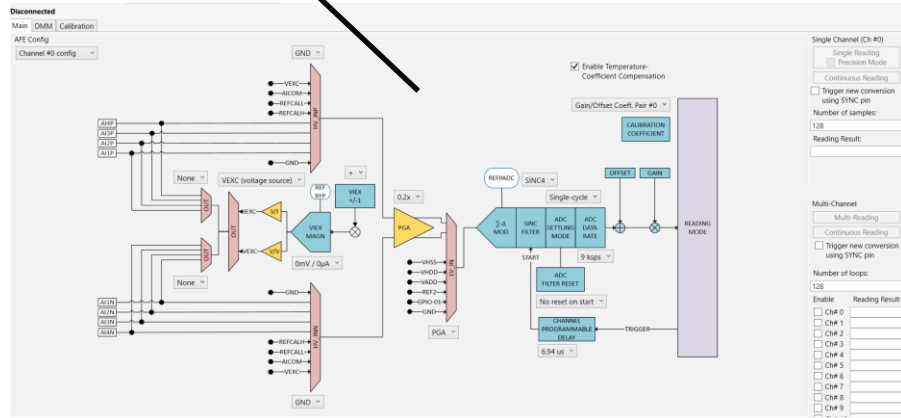
NAFE13388/NAFE11388 Board, to be connected through FRDM header to LPC MCU

NAFE Board

PC

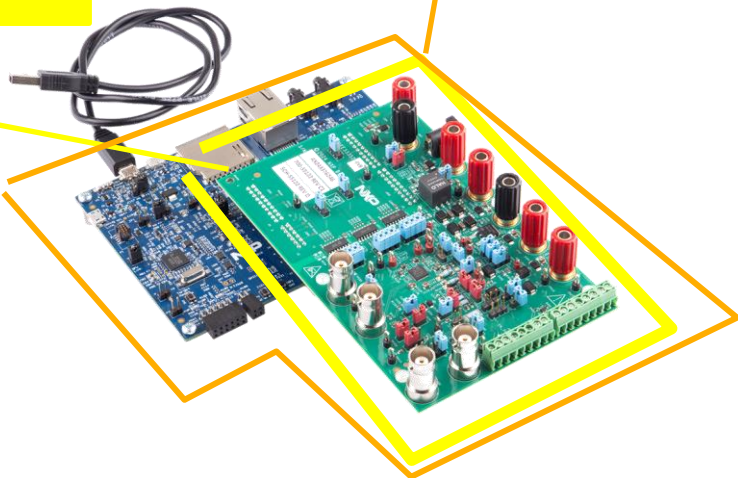
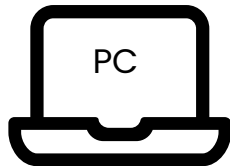
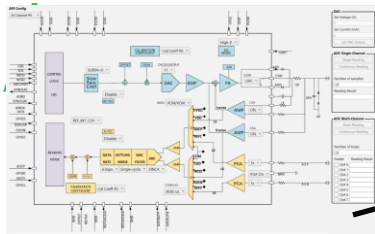
NAFE and MCU Board

GUI



KITNAFE13388-EVB KITNAFE11388-EVB

(NAFE Board + USB cable + MCU board + AFE SW Drivers)
Works with NXP GUI



KITNAFE33352-EVB

NAFE33352-EVB

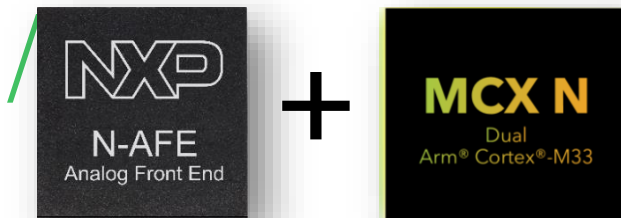
Connect or kit your board with NXP MCUs

- **NAFE33352-EVK**
 - NAFE board + USB cable + LPC54xx MCU + adaptor + AFE SW Drivers
 - Works with NXP GUI

Connect your board with SPI

- **NAFE33352-EVB**
 - NAFE + adaptor

Coming NEXT: **NAFE33352-UIOM**
Software Configurable Input and Output AFE
Arduino® Shield Board



NAFE FRDM UIM

NAFE13388-UIM

Enable compact hardware design and enhance software configurability with universal open-CMSIS driver packs



Compact Design

- Small 89mm x 50mm size
- Arduino UNO connector connects to any NXP MCU platform
- FRDM compatible

Enhanced Analog Sensing

- 8 software configurable analog inputs to measure Voltage, Current, RTD, and Thermocouple
- Embedded RTD for Cold Junction Compensation
- Current sense resistors controlled by GPIOs

Advanced Analog Features

- Voltage or current excitation sources
- Factory and User Calibration support
- Advanced Diagnostics and anomaly detection

Plug & Play

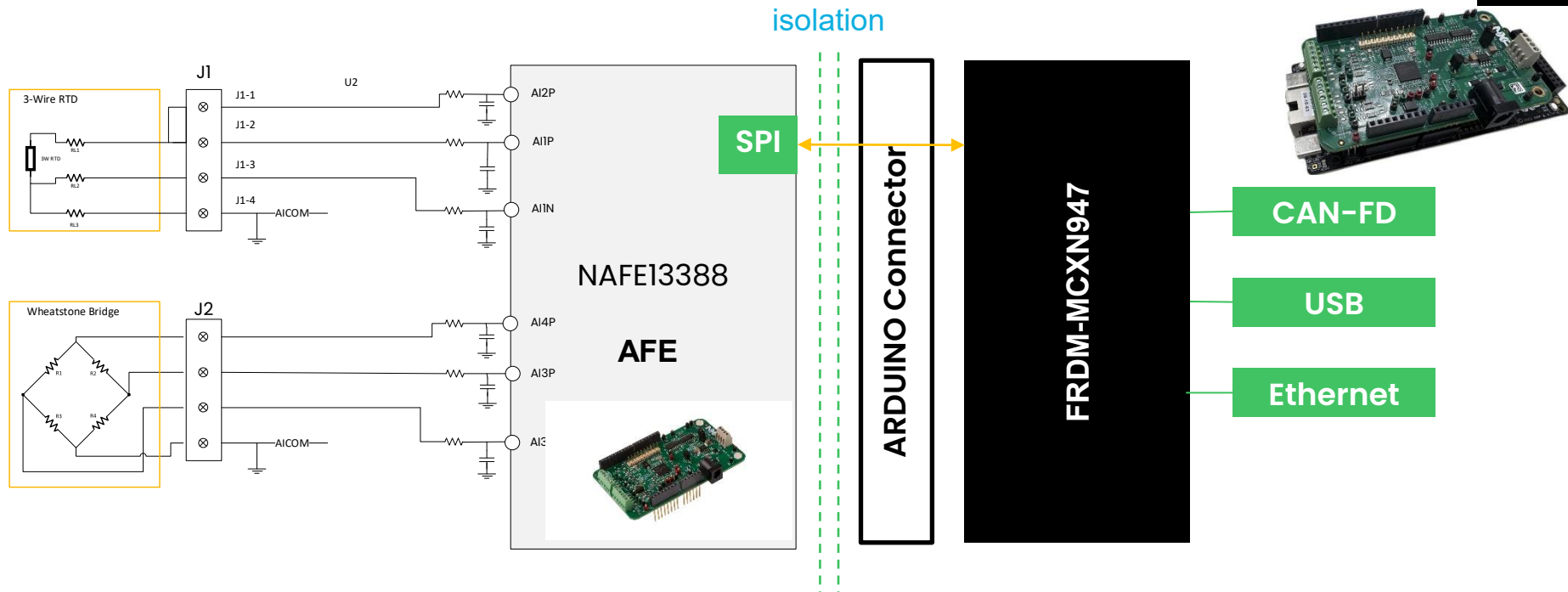
- NAFE GUI
- Code Examples



Backed By MCUXpresso | Unified Developer Experience



Application Board Overview



Accurate and Precise Sensor Data

Measures voltage, current, temperature, weight, pressure, resistance...

Robustness and Diagnostics

Wide range of diagnostic functions for anomaly detection and failure prediction

Edge Processing

MCX N947 dual Arm® Cortex®-M33 cores @ 150 MHz with 2 MB Flash, eIQ® Neutron NPU, DSP accelerator, and EdgeLock® Secure Enclave

Connectivity

CAN FD, 10/100 Mbps Ethernet or USB.

← → ↺ ↻

nxp.com/design/design-center/development-boards-and-designs/NAFE13388-UIM

🔍 ☆ 📄 🗨

NXP

PRODUCTS APPLICATIONS DESIGN CENTER SUPPORT COMPANY **STORE**

🔍 Search My NXP Account English ▾

Overview Product Details Documentation Design Resources ⓘ Training Support **BUY OPTIONS** GET STARTED



⌵

The NAFE13388-UIM 8-channel universal input AFE Arduino shield board is designed for the evaluation of the NAFExx388, which is a highly configurable, industrial-grade multichannel analog input analog front-end (AFE). This Arduino shield board showcases compact hardware design and enhanced software configurability with the universal open-CMSIS driver packs.

The shield board supports quick evaluation and plug-and-play compatibility with NXP's MCU boards (MCX-N and LPC) having an Arduino connector. Users can quickly perform voltage, current and temperature measurements (RTD and thermocouple), reducing the development time of the final application.

DESIGN FILES

SOFTWARE

Roll over image to zoom in



Product Details

[Block Diagram](#) | [Supported Devices](#) | [Features](#) | [Applications](#)

 Filter by keyword

4 software files

Sort by | Relevance 

BSPS AND DEVICE DRIVERS

NAFE13388 Universal Input Analog Front-End with Demo App (SDK)

EXTERNAL Rev 1 Oct 16, 2024 NAFE13388-UNIVERSAL-INPUT-DEMO-APP-SDK

DOWNLOAD



BSPS AND DEVICE DRIVERS

NAFE13388 Universal Input Analog Front-End - Application Code Hub

EXTERNAL Rev 1 Oct 16, 2024 NAFE13388-UNIVERSAL-INPUT-APPLICATION-CODE-HUB

DOWNLOAD



DEVELOPMENT IDES AND BUILD TOOLS

elQ® Time Series Studio for Edge AI Development

DOWNLOAD OPTIONS



SOFTWARE DEVELOPMENT RESOURCES

GUI for NAFE13388-UIM

ZIP Rev 1.0 May 15, 2025 2.19 MB GUI-NAFE13388-UIM

 **Account Required**

DOWNLOAD



← → ↺ 🏠 github.com/nxp-appcodehub/dm-nafe13388-application-examples 🔍 📄 ⭐

🐙 Platform ▾ Solutions ▾ Resources ▾ Open Source ▾ Enterprise ▾ Pricing 🔍 Sign in

📁 nxp-appcodehub / **dm-nafe13388-application-examples** Public 🔔 Notifications 🍴 Fork 0

<> Code ⓘ Issues 🏷 Pull requests 🛡 Security 📊 Insights

👤 main ▾ 🌿 1 Branch 🏷 0 Tags 🔍 Go to file <> Code ▾

👤 Anurag070303 frdm-mcxn947:nafe13388:Bug Fixes 🗨

📁 examples_CLI	frdm-mcxn947:nafe13388:Bug Fixes
📁 examples_CLI_CMSIS	frdm-mcxn947:nafe13388:Bug Fixes
📁 examples_GUI	frdm-mcxn947:nafe13388:Bug Fixes
📁 examples_GUI_CMSIS	frdm-mcxn947:nafe13388:Bug Fixes
📁 images	frdm-mcxn947:nafe13388:Bug Fixes
📄 LICENSE.txt	frdm-mcxn947:nafe13388:Bug Fixes 5 months ago
📄 README.md	frdm-mcxn947:nafe13388:Bug Fixes 5 months ago
📄 SW-Content-Register.txt	frdm-mcxn947:nafe13388:Bug Fixes 5 months ago
📄 dm-nafe13388-application-examples.xml	frdm-mcxn947: nafe13388-uim: complete cli and gui applic... 2 years ago

📄 Clone ⓘ

HTTPS GitHub CLI

<https://github.com/nxp-appcodehub/dm-nafe13388-application-examples> 📄

Clone using the web URL

🖱 Open with GitHub Desktop

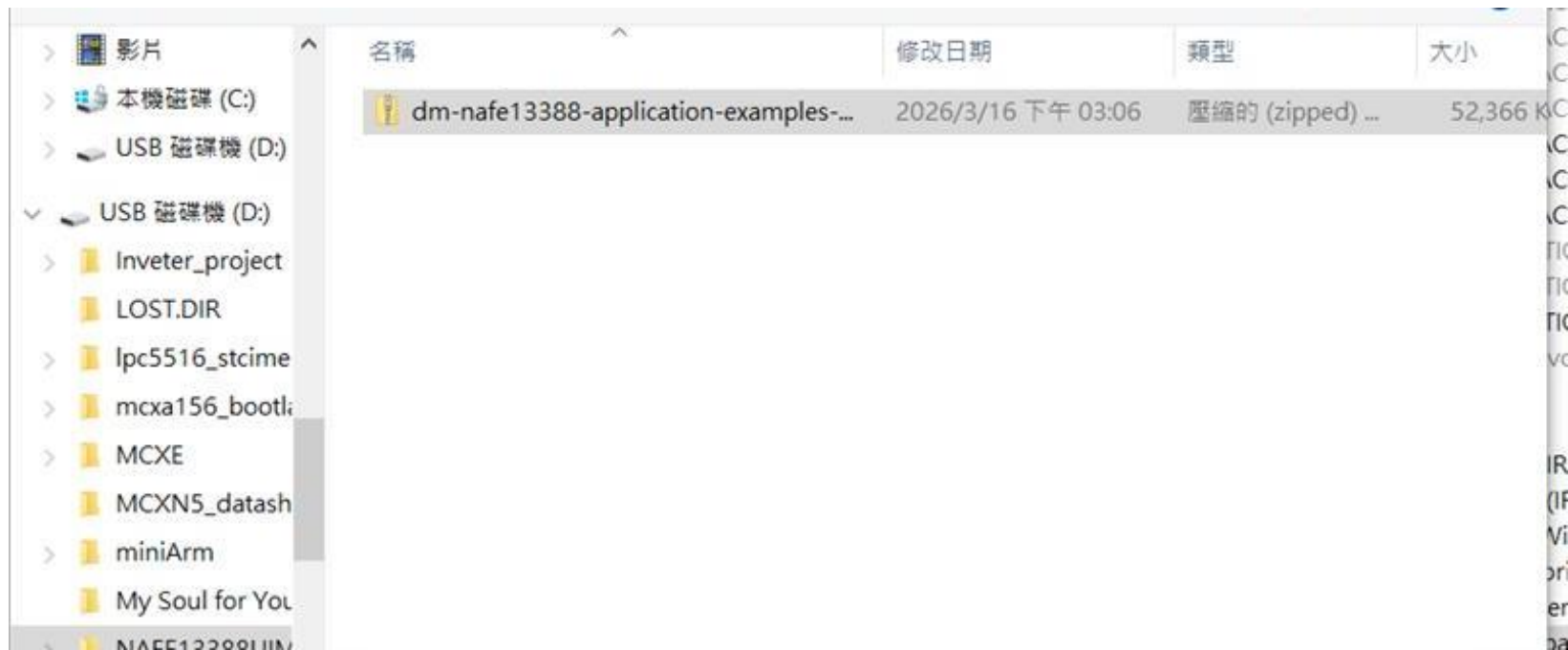
📄 Download ZIP

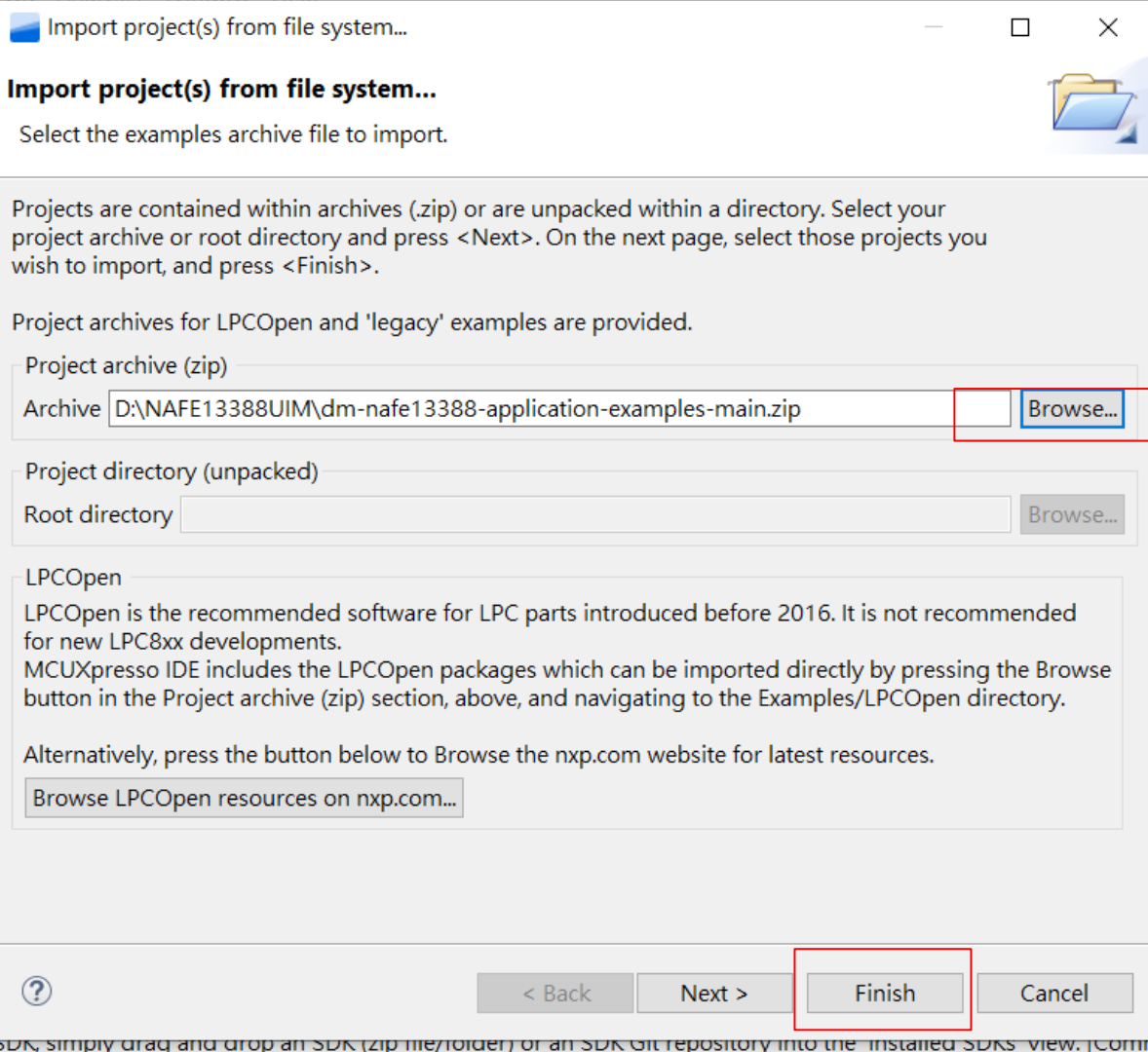
About

This is a demo application provides a command line interface for these 6 applications: Analog to Digital Converter, Current Sensing, Conversion Voltage, SCSR Conversion Voltage, using MCMR Conversion, Wires using SCCR Conversion, Scale

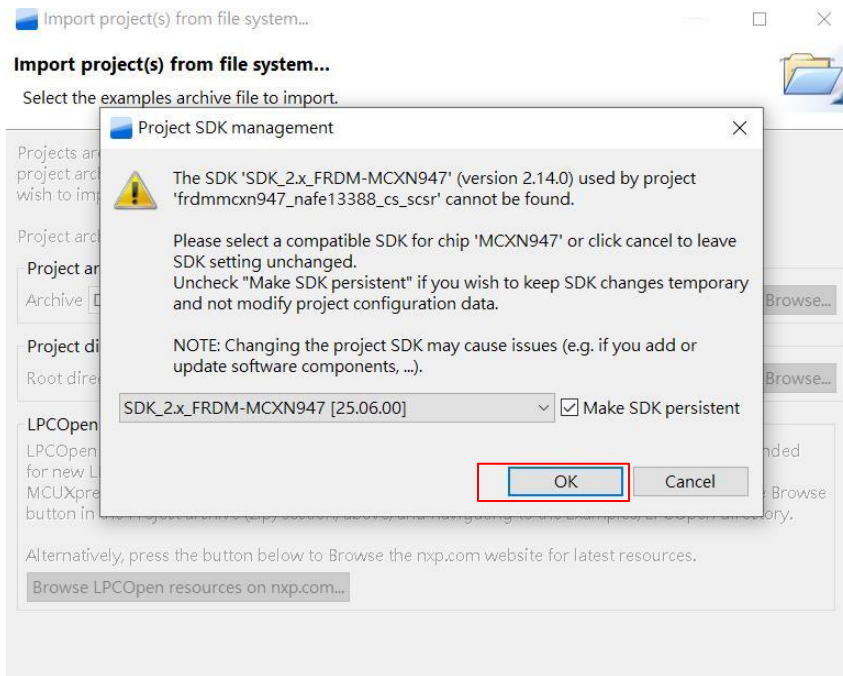
- 📖 Readme
- 📄 View license
- 🔍 Activity
- 📁 Custom properties
- ⭐ 1 star
- 👁 0 watching
- 🍴 0 forks
- 📄 Report repository

Releases





指到zip file



The screenshot displays the MCUXpresso IDE interface. The top menu bar includes File, Edit, Navigate, Search, Project, ConfigTools, Run, RTOS, Analysis, Window, and Help. The Project Explorer on the left shows a tree of files and folders, with 'MCUX947_DIB <Debug>' selected. The Quickstart Panel on the bottom left provides options for creating or importing a project, building the project, and debugging. The CDT Build Console on the bottom right shows the build output for the 'MCUX947_DIB' target, including memory usage and build time.

Project Explorer:

- frdm_mcx156_mcu_boot_opensource
- frdm_mcx156_wads_ota_mcu_boot_basics
- frdm_mcx947_nafe13388
- frdm_mcx947_nafe13388_cs_scsr
- frdm_mcx947_nafe13388_four_wire_rtd
- frdm_mcx947_nafe13388_two_wire_rtd
- frdm_mcx947_nafe13388_vs_mcmr
- frdm_mcx947_nafe13388_vs_scsr
- frdm_mcx947_nafe13388_weightScale
- lpcpresso55s16_ctimer_pwm_example
- lpcpresso55s16_ctimer_pwm_interrupt_example
- lpcpresso55s16_hello_world
- MCUX947_DIB <Debug>**
- Project Settings
- Binaries
- Includes
- CMSIS
- board
- component
- device
- drivers

Quickstart Panel:

MCUXpresso IDE Quickstart
Project: MCUX947_DIB [Debug]

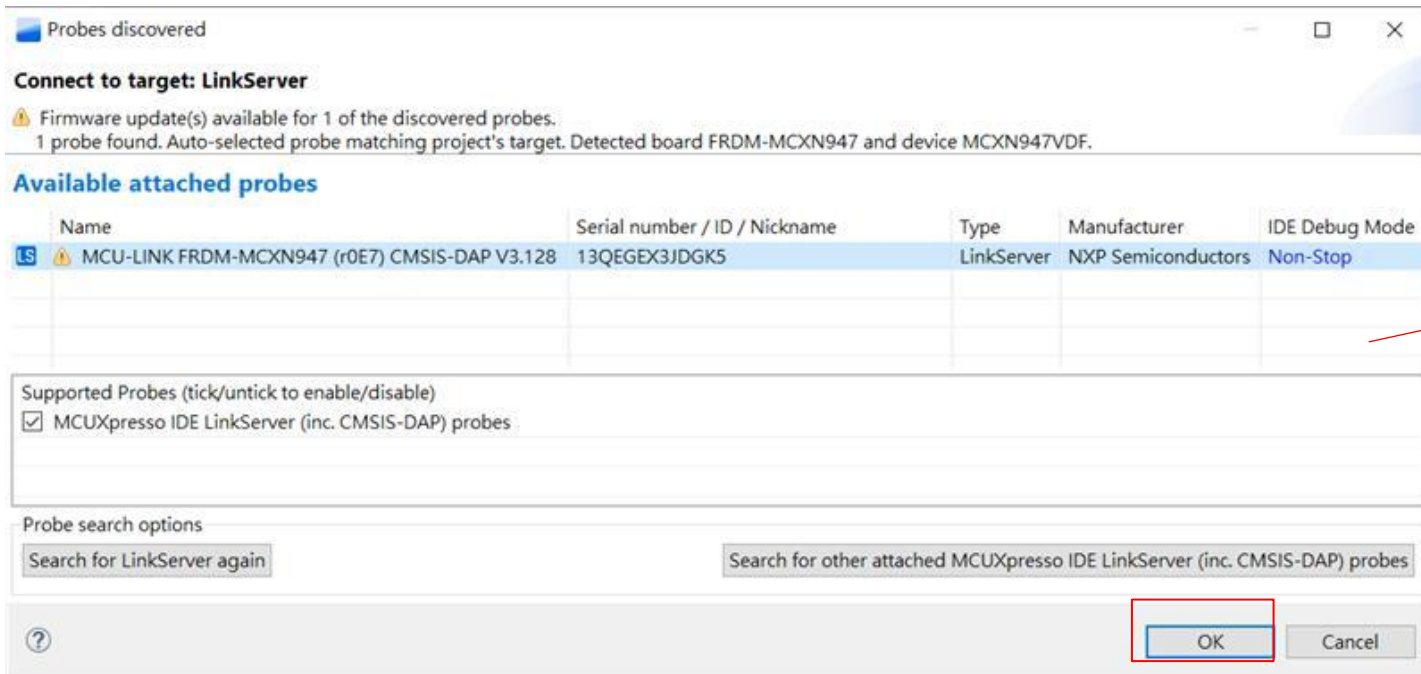
- Create or import a project
 - Create a new C/C++ project...
 - Import SDK example(s)...
 - Import from Application Code Hub...
 - Import project(s) from file system...
 - Import executable from file system...
- Build your project
 - Build
 - Clean
- Debug your project

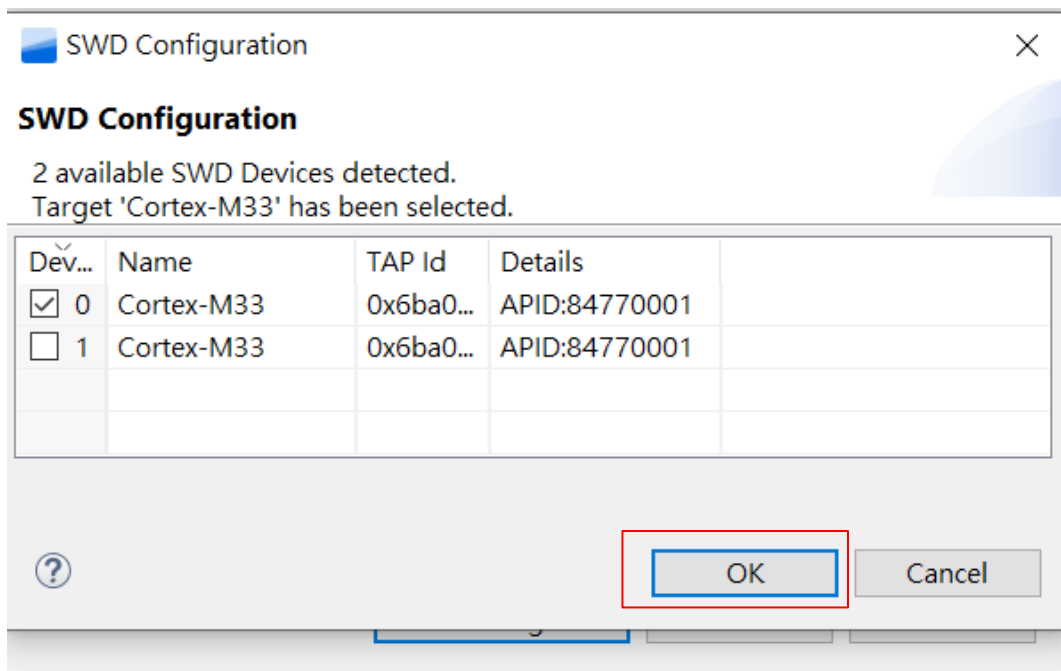
CDT Build Console [MCUX947_DIB]

```
SRAM: 0 B 32 KB 0.00%
SRAM: 0 B 32 KB 0.00%
USB_RAM: 0 B 4 KB 0.00%
Finished building target: MCUX947_DIB.axf

Performing post-build steps
arm-none-eabi-size "MCUX947_DIB.axf"; # arm-none-eabi-objcopy -v -O binary "MCUX947_DIB.axf" "MCUX947_DIB.bin"; # checksum
text data bss dec hex filename
64032 356 151340 215728 34ab0 MCUX947_DIB.axf

15:24:54 Build Finished. 0 errors, 0 warnings. (took 6s.721ms)
```





NAFE13388-UIM GUI v1.4.0
File Settings Help

NAFE13388

Ports Available: COM4 (2) Connect (3) Search (1)

Main Configurations GPIOs Configuration Calibration Register Map Application Example

Analog Input

CH0 (2)

Enable Temperature-Coefficient Compensation

Coefficient Pair 0

Single Channel Mode

Single Reading (SCSR) N samples

Continuous Reading (SCCR) 10

Average STD

Result 0.00000 0.00e+000

Multi Channel Mode

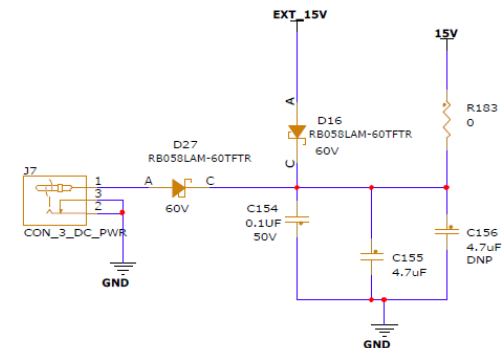
Multi Reading (MCMR) N samples

Continuous Reading (MCCR) 10

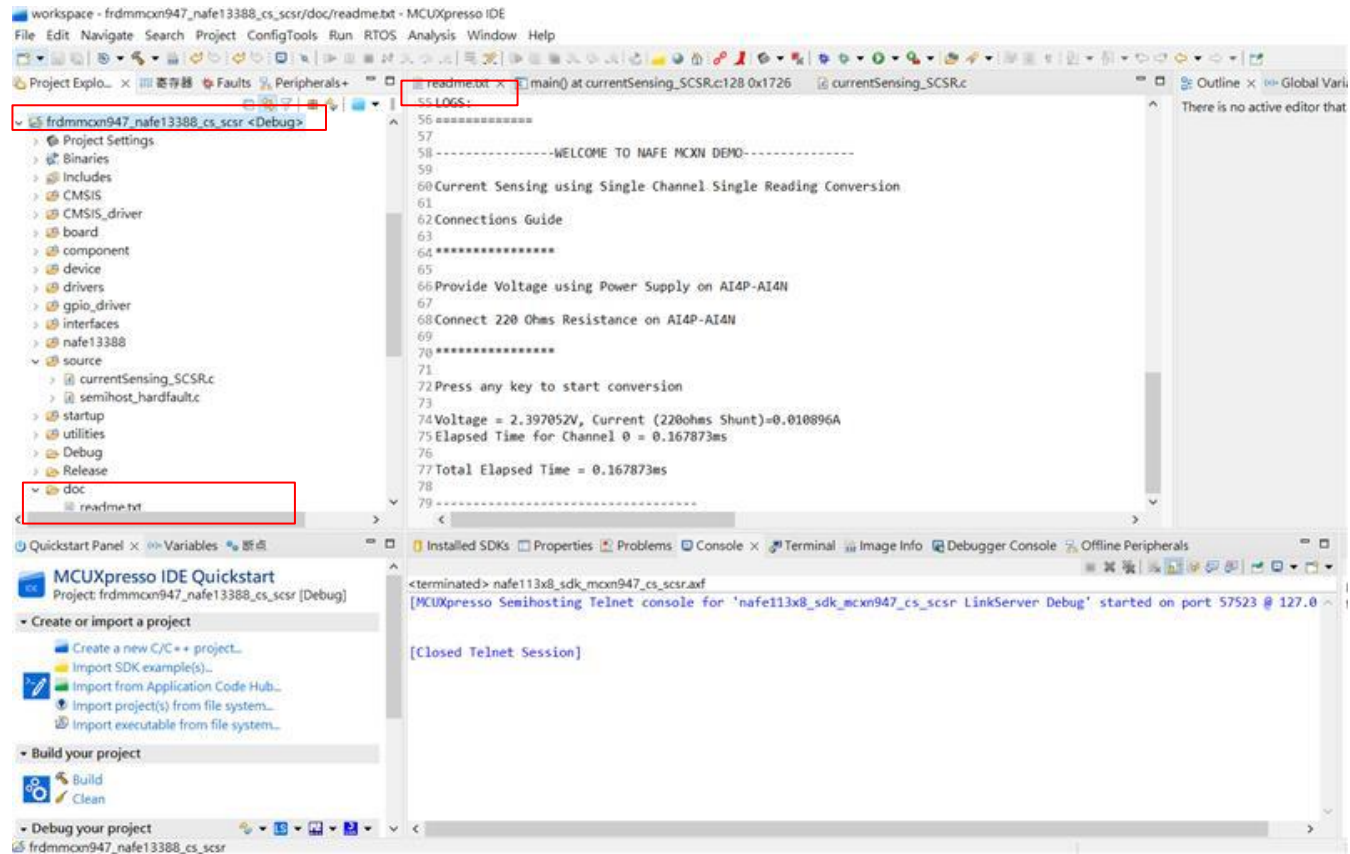
Results

CH	Average	STD
0	0.00000	0.00e+000
1	0.00000	0.00e+000
2	0.00000	0.00e+000
3	0.00000	0.00e+000
4	0.00000	0.00e+000
5	0.00000	0.00e+000
6	0.00000	0.00e+000
7	0.00000	0.00e+000
8	0.00000	0.00e+000
9	0.00000	0.00e+000
10	0.00000	0.00e+000
11	0.00000	0.00e+000
12	0.00000	0.00e+000
13	0.00000	0.00e+000
14	0.00000	0.00e+000
15	0.00000	0.00e+000





15V

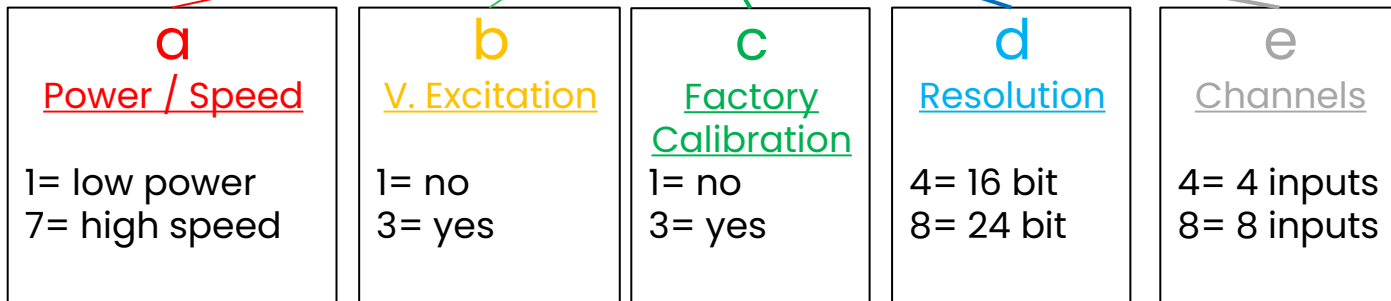


Questions

AVNET[®]



NAFE**a****b****c****d****e**B40BS



V/I Input

NAFE11388: **Low Power** – **No VIEX** – **Cal** – **24 bits** – 8 ch

NAFE71388: **High Speed** – **No VIEX** – **Cal** – **24 bits** – 8 ch

NAFE11348: **Low Power** – **No VIEX** – **Cal** – **16 bits** – 8 ch

V/I/RTD/TC Input

NAFE13388: **Low Power** – **VIEX** – **Cal** – **24 bits** – 8 ch

NAFE13188: **Low Power** – **VIEX** – **non Cal** – **24 bits** – 8 ch

NAFE73388: **High Speed** – **VIEX** – **Cal** – **24 bits** – 8 ch

NAFE13344: **Low Speed** – **VIEX** – **non Cal** – **16 bits** – 4 ch

Next to be released

Q4 2024

NAFE13148: **Low Power** – **VIEX** – **non Cal** – **16 bits** – 8 ch